

ACME Multimodal Content Moderation: Run Evidence

Author: Dennis O'Higgins. All runs were made in the Udacity workspace (Python 3.12.12) on 7 October 2026.

Contents: (1) unit test output, (2) eval run output (text, plus image as a supplementary run), (3) screenshot of a flagged message in the Gradio chat with its rationale, (4) screenshots of the traces in the Phoenix dashboard showing the span names.

1. Unit tests: `uv run pytest tests/ -vv --ignore=tests/test_env_setup.py`

```
===== test session starts =====
platform linux -- Python 3.12.12, pytest-8.4.2, pluggy-1.6.0 -- /voc/work/code/project/starter/.venv/bin/python
cachedir: .pytest_cache
rootdir: /voc/work/code/project/starter
configfile: pyproject.toml
plugins: anyio-4.11.0, logfire-4.13.2, asyncio-1.2.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collecting ... collected 45 items

tests/test_audio_agent.py::test_moderate_audio_exists PASSED [ 2%]
tests/test_audio_agent.py::test_moderate_audio_returns_audio_moderation_result PASSED [ 4%]
tests/test_audio_agent.py::test_moderate_audio_has_required_fields PASSED [ 6%]
tests/test_audio_agent.py::test_moderate_audio_rationale_not_empty PASSED [ 8%]
tests/test_audio_agent.py::test_moderate_audio_transcription_not_empty PASSED [ 11%]
tests/test_gradio_app.py::test_create_chat_interface_returns_blocks PASSED [ 13%]
tests/test_gradio_app.py::test_chatbot_uses_messages_type PASSED [ 15%]
tests/test_gradio_app.py::test_chatbot_has_copy_button PASSED [ 17%]
tests/test_gradio_app.py::test_demo_has_chatbot_component PASSED [ 20%]
tests/test_gradio_app.py::test_demo_has_multimodal_textbox PASSED [ 22%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_multiple_files PASSED [ 24%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_correct_file_types PASSED [ 26%]
tests/test_gradio_app.py::test_chat_with_gemini_calls_moderation PASSED [ 28%]
tests/test_gradio_app.py::test_chat_with_gemini_blocks_flagged_content PASSED [ 31%]
tests/test_image_agent.py::test_moderate_image_exists PASSED [ 33%]
tests/test_image_agent.py::test_moderate_image_returns_image_moderation_result PASSED [ 35%]
tests/test_image_agent.py::test_moderate_image_has_required_fields PASSED [ 37%]
tests/test_image_agent.py::test_moderate_image_rationale_not_empty PASSED [ 40%]
tests/test_moderation_result.py::TestModerationResult::test_has_rationale_field PASSED [ 42%]
tests/test_moderation_result.py::TestModerationResult::test_rationale_is_required PASSED [ 44%]
tests/test_moderation_result.py::TestModerationResult::test_is_pydantic_model PASSED [ 46%]
tests/test_moderation_result.py::TestTextModerationResult::test_has_all_required_fields PASSED [ 48%]
tests/test_moderation_result.py::TestTextModerationResult::test_field_types PASSED [ 51%]
tests/test_moderation_result.py::TestTextModerationResult::test_inherits_from_moderation_result PASSED [ 53%]
tests/test_moderation_result.py::TestTextModerationResult::test_all_fields_are_required PASSED [ 55%]
tests/test_moderation_result.py::TestImageModerationResult::test_has_all_required_fields PASSED [ 57%]
tests/test_moderation_result.py::TestImageModerationResult::test_field_types PASSED [ 60%]
tests/test_moderation_result.py::TestImageModerationResult::test_inherits_from_moderation_result PASSED [ 62%]
tests/test_moderation_result.py::TestImageModerationResult::test_all_fields_are_required PASSED [ 64%]
tests/test_moderation_result.py::TestVideoModerationResult::test_has_all_required_fields PASSED [ 66%]
tests/test_moderation_result.py::TestVideoModerationResult::test_field_types PASSED [ 68%]
tests/test_moderation_result.py::TestVideoModerationResult::test_inherits_from_moderation_result PASSED [ 71%]
tests/test_moderation_result.py::TestVideoModerationResult::test_all_fields_are_required PASSED [ 73%]
tests/test_moderation_result.py::TestAudioModerationResult::test_has_all_required_fields PASSED [ 75%]
tests/test_moderation_result.py::TestAudioModerationResult::test_field_types PASSED [ 77%]
tests/test_moderation_result.py::TestAudioModerationResult::test_inherits_from_moderation_result PASSED [ 80%]
tests/test_moderation_result.py::TestAudioModerationResult::test_all_fields_are_required PASSED [ 82%]
tests/test_text_agent.py::test_moderate_text_exists PASSED [ 84%]
tests/test_text_agent.py::test_moderate_text_returns_text_moderation_result PASSED [ 86%]
tests/test_text_agent.py::test_moderate_text_has_required_fields PASSED [ 88%]
tests/test_text_agent.py::test_moderate_text_rationale_not_empty PASSED [ 91%]
tests/test_video_agent.py::test_moderate_video_exists PASSED [ 93%]
tests/test_video_agent.py::test_moderate_video_returns_video_moderation_result PASSED [ 95%]
tests/test_video_agent.py::test_moderate_video_has_required_fields PASSED [ 97%]
tests/test_video_agent.py::test_moderate_video_rationale_not_empty PASSED [100%]
===== 45 passed in 34.91s =====
```

2. Eval run: uv run evals/text/test_cases.py (EVAL_NUM_REPEATS=1)

The text suite scored 91.7%. The only failed assertion is the LLM judge on unfriendly_text: the agent set the correct flags (is_unfriendly=True, is_unprofessional=True) but wrote a truncated rationale, so the judge marked the rationale as insufficient. A score below 100% is expected because both the agent and the judge are language models.

Evaluating run_text_moderation 100% 0:00:00
Evaluation Summary: run_text_moderation

Case ID	Inputs	Outputs	Assertions
professional_text	[TextInput(text_file='/voc/work/code/project/starter/evals/test_data/professional_text.txt')]	rationale='The message is friendly and professional. It offers help, provides clear instructions, and uses polite language. No PII was detected.' contains_pii=False is_unfriendly=False is_unprofessional=False	✓✓✓✓
text_with_pii	[TextInput(text_file='/voc/work/code/project/starter/evals/test_data/text_with_pii.txt')]	rationale='The message contains PII such as name, address, email, phone number, and account number. It is friendly and professional.' contains_pii=True is_unfriendly=False is_unprofessional=False	✓✓✓✓
unfriendly_text	[TextInput(text_file='/voc/work/code/project/starter/evals/test_data/unfriendly_text.txt')]	rationale='The customer representative' contains_pii=False is_unfriendly=True is_unprofessional=True	✓x✓✓
Averages			91.7% ✓

Supplementary eval run: uv run evals/image/test_cases.py (100%)

Evaluating run_image_moderation 100% 0:00:00
Evaluation Summary: run_image_moderation

Case ID	Inputs	Outputs	Assertions
professional_image	[ImageInput(image_file='/voc/work/code/project/start er/evals/test_data/professional_image.jpg')]	rationale='The image is clear, professional, and suitable for a customer service setting. It does not contain any disturbing content or personally-identifiable information.' contains_pii=False is_disturbing=False is_low_quality=False	✓✓✓✓
image_with_person	[ImageInput(image_file='/voc/work/code/project/start er/evals/test_data/image_with_person.jpg')]	rationale='The image contains a person, which is considered personally-identifiable information (PII).'	✓✓✓✓
low_quality_image	[ImageInput(image_file='/voc/work/code/project/start er/evals/test_data/low_quality_image.jpg')]	rationale='The image contains a hand, which is considered PII. The image is also slightly blurry.' contains_pii=True is_disturbing=False is_low_quality=True	✓✓✓✓
Averages			100.0% ✓

3. Gradio chat: a message flagged by moderation, with the rationale as displayed

The trainee's message "Stop whining. That is your problem, not mine. Maybe read the manual next time." was flagged as unfriendly and unprofessional. It was not sent to the customer agent; the chat shows the blocked notice and the Moderation Agent Feedback panel shows the rationale.

 **ACME Customer Service Training Agent**

Welcome to ACME Corporation's customer service training!

Chatbot

My ACME Power Widget Pro just DIED. No warning, no explanation, just DEAD. This is unacceptable! I want my money back, NOW!

I am sorry but I absolutely cannot offer a refund

Are you KIDDING me?! You can't offer a refund for a product that completely failed?! What kind of operation are you running here?!

Stop whining. That is your problem, not mine. Maybe read the manual next time.

[This content was flagged by moderation and not sent to the AI. Please try again.]







Moderation Agent Feedback

 **Content flagged:** Content flagged: The customer representative used dismissive and rude language ("Stop whining") and shifted blame to the customer ("That is your problem, not mine."). The suggestion to "read the manual next time" was delivered in a condescending manner rather than being a helpful suggestion. This tone is both unfriendly and unprofessional.

 **End Conversation**

 **Chat Guidelines**

The AI acts as a customer complaining about an ACME product. Try to resolve the customer's issue. You can type messages, upload images/videos, or record audio.

 **Content Moderation**

All messages and media are automatically checked for:

- Inappropriate content
- Personally identifiable information
- Unprofessional language

4. Phoenix traces: span names

All spans of the session: moderate_text, chat_turn, llm_customer and the Pydantic AI agent and LLM spans.

PHOENIX

Projects

Datasets & Experiments

Playground

Prompts

APIs

Star on GitHub 12k

Settings

Documentation

Support

Dark

Projects > default

Total Traces1

Total Cost<\$0.01

Latency P50🕒 459.80s

Latency P99🕒 459.80s

Stream🔴

Last 7 Days📅

Spans

Traces

Sessions

Metrics

Config

filter condition (e.x. span_kind == 'LLM')

+

Root Spans

All

Columns📄

🔍

	status	kind	name	input	output	annotations ⓘ	start time
☐	🕒	unknown	moderate_text	--	--		07/10/2026, 16:08
☐	🕒	unknown	chat_turn	Stop whining. That is ...	--		07/10/2026, 16:08
☐	🕒	llm	chat gemini-2.5-flash-lite.	I am sorry but I absolu...	--		07/10/2026, 16:07
☐	🕒	agent	agent run	This is the next messa...	Are you KIDDING me?!...		07/10/2026, 16:07
☐	🕒	unknown	llm_customer	--	--		07/10/2026, 16:07
☐	🕒	unknown	moderate_text	--	--		07/10/2026, 16:07
☐	🕒	unknown	chat_turn	I am sorry but I absolu...	Are you KIDDING me?!...		07/10/2026, 16:07
☐	🕒	llm	chat gemini-2.5-flash-lite.	Welcome to ACME Cu...	--		07/10/2026, 16:05
☐	🕒	agent	agent run	This is the next messa...	My ACME Power Widg...		07/10/2026, 16:05
☐	🕒	unknown	llm_customer	--	--		07/10/2026, 16:05
☐	🕒	unknown	moderate_text	--	--		07/10/2026, 16:05
☐	🕒	unknown	chat_turn	Welcome to ACME Cu...	My ACME Power Widg...		07/10/2026, 16:05
☐	🕒	llm	chat gemini-2.5-flash-lite.	This is the next messa...	--		07/10/2026, 16:03

4 (continued). Root conversation span with session.id, and the nested chat_turn, moderate_text and llm_customer spans

The screenshot displays the Phoenix AI interface, specifically the 'Trace Details' view. The left sidebar contains navigation links: Projects, Datasets & Experiments, Playground, Prompts, APIs, Star on GitHub, Settings, Documentation, Support, and Dark mode. The main content area is titled 'Trace Details' and shows a tree view of the trace. The root node is 'conversation', which contains several 'chat_turn' nodes. Each 'chat_turn' node has sub-nodes for 'moderate_text', 'llm_customer', and 'agent run'. The 'agent run' nodes are further detailed with 'chat gemini-2.5-...' and a token count (e.g., 183, 237, 268). The top of the interface shows 'Trace Status' as 'UNSET', 'Total Cost' as '<\$0.01', and 'Latency' as '924.13s'. The right sidebar contains sections for 'Annotation Summary', 'Edit Annotations', and 'Notes'. The 'Edit Annotations' section shows a message: 'No annotation configurations for this project' with a link to 'Configure Annotation Configs'. The 'Notes' section has a text input field labeled 'Add a note' and a send button.

PHOENIX

Trace Details

Trace Status: UNSET, Total Cost: <\$0.01, Latency: 924.13s

Trace: conversation

chat_turn

- moderate_text
- llm_customer
 - agent run
 - chat gemini-2.5-... (183)

chat_turn

- llm_customer
 - agent run
 - chat gemini-2.5-... (237)

chat_turn

- llm_customer
 - agent run
 - chat gemini-2.5-... (268)

chat_turn

- moderate_text
- llm_customer

unknown conversation

924.13s at 10/7/2026 15:58:54 PM

Info Annotations 0 Attributes Events 0

All Attributes

```
1 {
2   "session": {
3     "id": "c5b019c1-3935-476c-951b-3c393b124fc4"
4   }
5 }
```

Annotation Summary 5

Edit Annotations E

+ Annotation

No annotation configurations for this project

Configure Annotation Configs

Notes N

Add a note

4 (continued). The flagged chat_turn span with its feedback, moderation.flagged and session.id attributes

The screenshot displays the Phoenix AI interface, specifically the 'Trace Details' view. The left sidebar contains navigation links: Projects, Datasets & Experiments, Playground, Prompts, APIs, Star on GitHub, Settings, Documentation, Support, and Dark mode. The main area is divided into three sections: Trace, Attributes, and Annotations.

Trace Section: Shows a tree view of the trace. The selected trace is 'chat_turn' (unknown) at 10/7/2026 16:08:11 PM. It has a total cost of <\$0.01 and a latency of 459.80s. The trace status is UNSET. The trace tree shows a sequence of spans: 'chat_turn' (183), 'moderate_text', 'llm_customer', 'agent run', 'chat gemini-2.5-fla...' (237), 'chat_turn' (268), 'moderate_text', 'llm_customer', and 'agent run'.

Attributes Section: Displays the 'All Attributes' for the selected span. The attributes are as follows:

```
1 {
2   "history": {
3     "message_count": 4
4   },
5   "feedback": "⚠ Content flagged: Content flagged: The customer
representative used dismissive and rude language (\\"Stop whining\\") and
shifted blame to the customer (\\"That is your problem, not mine.\"). The
suggestion to \\"read the manual next time\\" was delivered in a
condescending manner rather than being a helpful suggestion. This tone is
both unfriendly and unprofessional.",
6   "moderation": {
7     "flagged": true
8   },
9   "input": {
10    "value": "Stop whining. That is your problem, not mine. Maybe read
the manual next time.",
11    "file_count": 0
12  },
13  "session": {
14    "id": "c5b019c1-3935-476c-951b-3c393b124fc4"
15  }
16 }
```

Annotations Section: Shows 'Annotation Summary' (5) and 'Edit Annotations' (E). There is a '+ Annotation' button. Below this, it states 'No annotation configurations for this project' with a link to 'Configure Annotation Configs'. At the bottom, there is a 'Notes' section (N) with an 'Add a note' button and a send icon.